

```
In[1]:= Clear["Global`*"];
```

0. Readme

1. (x', y', z') are represented by $(x3, y3, z3)$.
2. $V_{\lambda\lambda\lambda}^s, V_{\lambda\lambda\phi}^s, V_{\lambda\lambda r}^s, V_{\lambda\phi r}^s, V_{\phi\phi\lambda}^s, V_{\phi\phi\phi}^s, V_{\phi\phi r}^s, V_{rr\lambda}^s, V_{rr\phi}^s$, and V_{rrr}^s in Table 2 are represented by SphericalV $\lambda\lambda\lambda$, SphericalV $\lambda\lambda\phi$, SphericalV $\lambda\lambda r$, SphericalV $\lambda\phi r$, SphericalV $\phi\phi\lambda$, SphericalV $\phi\phi\phi$, SphericalV $\phi\phi r$, SphericalV $rr\lambda$, SphericalV $rr\phi$, and SphericalV rrr .
3. $V_{xxx}^c, V_{xxy}^c, V_{xxz}^c, V_{xyz}^c, V_{yyx}^c, V_{yyy}^c, V_{yyz}^c, V_{zzx}^c, V_{zzy}^c$, and V_{zzz}^c in Table 3 are represented by CartesianV xxx , CartesianV xxy , CartesianV xxz , CartesianV xyz , CartesianV yyx , CartesianV yyy , CartesianV yyz , CartesianV zzx , CartesianV zzy , and CartesianV zzz .
4. The integral kernels are considered and the complete expressions should be presented as $G\rho \int_{\lambda_1}^{\lambda_2} \int_{\phi_1}^{\phi_2} \int_{u_1}^{u_2} \kappa(\text{Kernel}) d u' d \phi' d \lambda'$.

1. Basic settings

```
In[2]:= A = x3 * Sin[λ] - y3 * Cos[λ];
        B = x3 * Cos[λ] + y3 * Sin[λ];

In[4]:= x = r * Cos[φ] * Cos[λ];
        y = r * Cos[φ] * Sin[λ];
        z = r * Sin[φ];

In[7]:= Δx = x - x3;
        Δy = y - y3;
        Δz = z - z3;

In[10]:= dx = -Sin[λ] * Δx + Cos[λ] * Δy;
         dy = -Sin[φ] * Cos[λ] * Δx - Sin[φ] * Sin[λ] * Δy + Cos[φ] * Δz;
         dz = Cos[φ] * Cos[λ] * Δx + Cos[φ] * Sin[λ] * Δy + Sin[φ] * Δz;

In[13]:= l = Sqrt[r^2 + x3^2 + y3^2 + z3^2 - 2 * r * (z3 * Sin[φ] + B * Cos[φ])];
```

2. GC expressions in spherical integral kernels from Table 2

```
In[14]:= SphericalVλλλ =
        3 * A * (6 * r^2 + x3^2 + y3^2 + 6 * z3^2 + 5 * (x3^2 - y3^2) * Cos[2 * λ] - 12 * r * z3 * Sin[φ] -
        12 * r * B * Cos[φ] + 10 * x3 * y3 * Sin[2 * λ]) / (2 * l^7);

In[15]:= SphericalVλλφ = -3 * (z3 * Cos[φ] - B * Sin[φ]) *
        (2 * r^2 - 3 * (x3^2 + y3^2) + 2 * z3^2 + 5 * (x3^2 - y3^2) * Cos[2 * λ] -
        4 * r * z3 * Sin[φ] - 4 * r * B * Cos[φ] + 10 * x3 * y3 * Sin[2 * λ]) / (2 * l^7);
```

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In[16]:= SphericalV $\lambda$  $\lambda$ r =
  3 * (r * Cos[2 *  $\phi$ ] * (x32 + y32 - 2 * z32 + (x32 - y32) * Cos[2 *  $\lambda$ ] + 2 * x3 * y3 * Sin[2 *  $\lambda$ ]) +
    B * Cos[ $\phi$ ] * (-6 * r2 + 3 * (x32 + y32) - 2 * z32 - 5 * (x32 - y32) * Cos[2 *  $\lambda$ ] +
      8 * r * z3 * Sin[ $\phi$ ] - 10 * x3 * y3 * Sin[2 *  $\lambda$ ]) +
    2 * r * (r2 - x32 - y32 + 2 * z32 + 3 * (x32 - y32) * Cos[2 *  $\lambda$ ] + 6 * x3 * y3 * Sin[2 *  $\lambda$ ]) -
    z3 * Sin[ $\phi$ ] * (6 * r2 - 3 * (x32 + y32) + 2 * z32 +
      5 * (x32 - y32) * Cos[2 *  $\lambda$ ] + 10 * x3 * y3 * Sin[2 *  $\lambda$ ])) / (2 * l7);

In[17]:= SphericalV $\lambda$  $\phi$ r = 15 * A * Cos[ $\phi$ ] * (r - z3 * Sin[ $\phi$ ] - B * Cos[ $\phi$ ]) * (z3 - B * Tan[ $\phi$ ]) / l7;

In[18]:= SphericalV $\phi$  $\phi$  $\lambda$  =
  3 * A * (r2 + x32 + y32 + z32 - 5 * z32 * (Cos[ $\phi$ ])2 - 2 * r * z3 * Sin[ $\phi$ ] - 2 * r * B * Cos[ $\phi$ ] +
    5 * z3 * B * Sin[2 *  $\phi$ ] - 5 * B2 * (Sin[ $\phi$ ])2) / l7;

In[19]:= SphericalV $\phi$  $\phi$  $\phi$  =
  3 * (z3 * Cos[ $\phi$ ] - B * Sin[ $\phi$ ]) * (-3 * (r2 + x32 + y32 + z32) + 5 * z32 * (Cos[ $\phi$ ])2 +
    6 * r * z3 * Sin[ $\phi$ ] + 6 * r * B * Cos[ $\phi$ ] - 5 * z3 * B * Sin[2 *  $\phi$ ] + 5 * B2 * (Sin[ $\phi$ ])2) / l7;

In[20]:= SphericalV $\phi$  $\phi$ r = 3 * (5 * z32 * B * (Cos[ $\phi$ ])3 + 5 * z3 * B2 * (Sin[ $\phi$ ])3 -
  z3 * Sin[ $\phi$ ] * (3 * r2 + x32 + y32 + z32 + 5 * y3 * z3 * Sin[2 *  $\phi$ ] * Sin[ $\lambda$ ]) +
  r * (r2 + x32 + y32 + z32 + 7 * z3 * B * Sin[2 *  $\phi$ ]) + r * (Sin[ $\phi$ ])2 * (2 * z32 - 5 * B2) +
  (Cos[ $\phi$ ])2 * (5 * z32 * (-r + z3 * Sin[ $\phi$ ]) + 2 * B2 * (r - 5 * z3 * Sin[ $\phi$ ])) -
  Cos[ $\phi$ ] * (-5 * x33 * (Sin[ $\phi$ ])2 * (Cos[ $\lambda$ ])3 -
    15 * x32 * y3 * (Sin[ $\phi$ ])2 * Sin[ $\lambda$ ] * (Cos[ $\lambda$ ])2 + y3 * Sin[ $\lambda$ ] *
    (3 * r2 + x32 + y32 + z32 - 5 * y32 * (Sin[ $\phi$ ])2 * (Sin[ $\lambda$ ])2) + x3 * Cos[ $\lambda$ ] *
    (3 * r2 + x32 + y32 + z32 + 5 * (Sin[ $\phi$ ])2 * (2 * z32 - 3 * y32 * (Sin[ $\lambda$ ])2))) / l7;

In[21]:= SphericalVrr $\lambda$  = -3 * A * (4 * r2 - x32 - y32 - z32 +
  (z3 * Sin[ $\phi$ ] + B * Cos[ $\phi$ ]) * (-8 * r + 5 * z3 * Sin[ $\phi$ ] + 5 * B * Cos[ $\phi$ ])) / l7;

In[22]:= SphericalVrr $\phi$  = -3 * (-z3 * Cos[ $\phi$ ] + B * Sin[ $\phi$ ]) * (4 * r2 - x32 - y32 - z32 +
  (z3 * Sin[ $\phi$ ] + B * Cos[ $\phi$ ]) * (-8 * r + 5 * z3 * Sin[ $\phi$ ] + 5 * B * Cos[ $\phi$ ])) / l7;

In[23]:= SphericalVrrr =
  -3 * (5 * (r - z3 * Sin[ $\phi$ ] - B * Cos[ $\phi$ ])3 - 3 * l2 * (r - z3 * Sin[ $\phi$ ] - B * Cos[ $\phi$ ])) / l7;

```

3. GC expressions in Cartesian integral kernels from Table 3

```

In[24]:= CartesianVxxx = FullSimplify[-3 * dx * (5 * dx2 - 3 * l2) / l7];

In[25]:= CartesianVxxy = FullSimplify[-3 * dy * (5 * dx2 - l2) / l7];

In[26]:= CartesianVxxz = FullSimplify[-3 * dz * (5 * dx2 - l2) / l7];

In[27]:= CartesianVxyz = FullSimplify[-15 * dx * dy * dz / l7];

In[28]:= CartesianVyyx = FullSimplify[-3 * dx * (5 * dy2 - l2) / l7];

In[29]:= CartesianVyyy = FullSimplify[-3 * dy * (5 * dy2 - 3 * l2) / l7];

In[30]:= CartesianVyyz = FullSimplify[-3 * dz * (5 * dy2 - l2) / l7];

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In[31]:= CartesianVzzx = FullSimplify[-3 * dx * (5 * dz^2 - l^2) / l^7];
In[32]:= CartesianVzzy = FullSimplify[-3 * dy * (5 * dz^2 - l^2) / l^7];
In[33]:= CartesianVzzz = FullSimplify[-3 * dz * (5 * dz^2 - 3 * l^2) / l^7];

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4. Differences between GC expressions in spherical and Cartesian integral kernels in Tables 2 and 3

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In[34]:= FullSimplify[SphericalVλλλ - CartesianVxxx]
Out[34]=
0

In[35]:= FullSimplify[SphericalVλλφ - CartesianVxxy]
Out[35]=
0

In[36]:= FullSimplify[SphericalVλλr - CartesianVxxz]
Out[36]=
0

In[37]:= FullSimplify[SphericalVλφr - CartesianVxyz]
Out[37]=
0

In[38]:= FullSimplify[SphericalVφφλ - CartesianVyyx]
Out[38]=
0

In[39]:= FullSimplify[SphericalVφφφ - CartesianVyyy]
Out[39]=
0

In[40]:= FullSimplify[SphericalVφφr - CartesianVyyz]
Out[40]=
0

In[41]:= FullSimplify[SphericalVrrλ - CartesianVzzx]
Out[41]=
0

In[42]:= FullSimplify[SphericalVrrφ - CartesianVzzy]
Out[42]=
0

In[43]:= FullSimplify[SphericalVrrr - CartesianVzzz]
Out[43]=
0

In[44]:= NotebookSave[EvaluationNotebook[]];

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